Group Policy Theory

Theory: What is Group Policy?

Group Policy is a feature of Windows Server Active Directory that lets an administrator control settings for many users and computers from one central place. Settings are saved in Group Policy Objects (GPOs). A GPO is linked to a site, a domain or an Organizational Unit (OU), and its settings are applied to the users and computers inside that container.

GPOs are used to enforce security rules (password and account lockout policies), restrict what users can change (for example blocking Control Panel on college lab or cybercafe PCs), and keep configuration consistent across all machines. They are created and managed in the Group Policy Management Console (GPMC) and edited in the Group Policy Management Editor.

Lab environment

The lab uses the domain lab.com. The EventPlanners OU holds the test users (msdhohni, Srikant m and virat k) and a global security group named FestTicketManagers. The user srikant.m was used on the client PC for all tests.

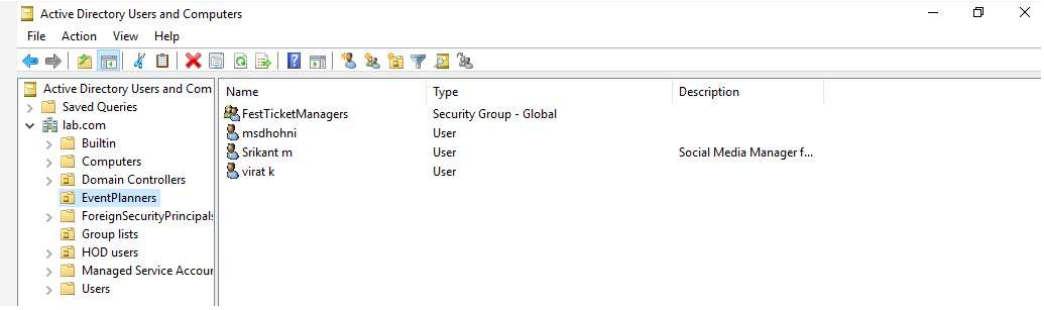


Figure 1: Active Directory Users and Computers showing the EventPlanners OU and its users

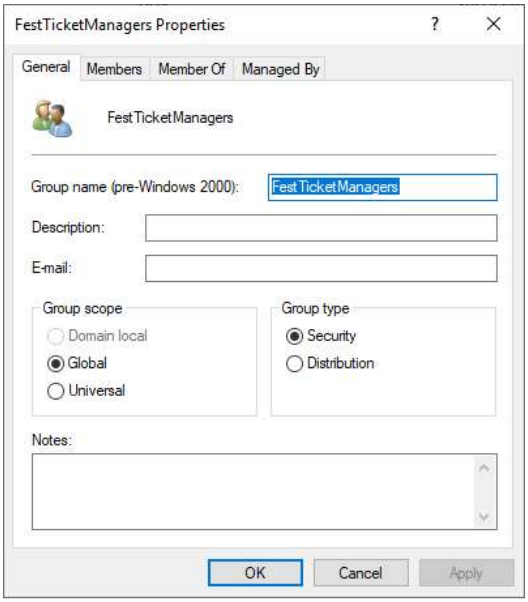


Figure 2: FestTicketManagers properties: Global scope, Security group type

Task 1: Open Group Policy Management Console (GPMC) on your Windows Server and create a new GPO named 'PasswordPolicyDemo' that enforces a minimum password length of 8 characters. Link it to your test OU and verify the setting is applied using gpresult.

Steps performed

1. Opened Server Manager > Tools > Group Policy Management.
2. Right-clicked Group Policy Objects, chose New, and named the GPO PasswordPolicyDemo.
3. Edited the GPO and went to Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy. Set Minimum password length to 8 characters.

4. Linked the GPO. Password and account lockout policies only take effect for domain accounts when they are applied at the domain level, so the link was made on lab.com (Figure 3). The GPO then applies to the users in the EventPlanners OU, including srikant.m.
5. On the client PC, signed in as srikant.m and ran the commands below to refresh and verify the policy.

```
gpupdate /force
gpresult /r
```

Result and verification

PasswordPolicyDemo is listed as a linked, enabled GPO on lab.com (Figure 3). The gpresult output shows PasswordPolicyDemo under Applied Group Policy Objects for CN=Srikantm, OU=EventPlanners, DC=lab,DC=com, so the policy is applied to the user (Figure 4). When the account password had to be changed, Windows showed the "password must be changed before signing in" message and the password-change form (Figures 5 and 6); the new password has to satisfy the minimum length of 8 characters.

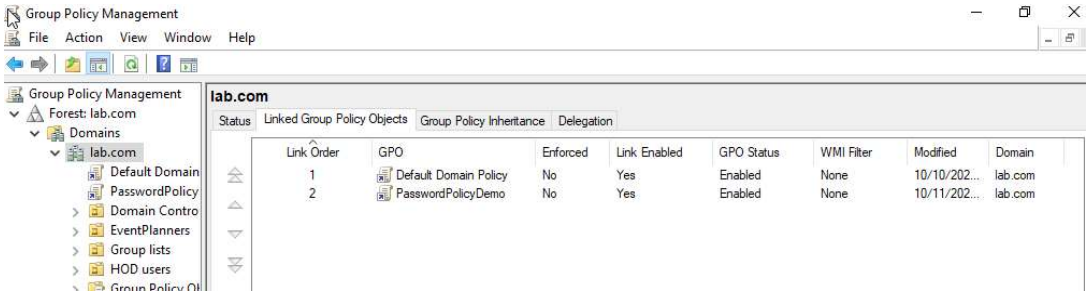


Figure 3: Linked Group Policy Objects on lab.com: Default Domain Policy and PasswordPolicyDemo (enabled)

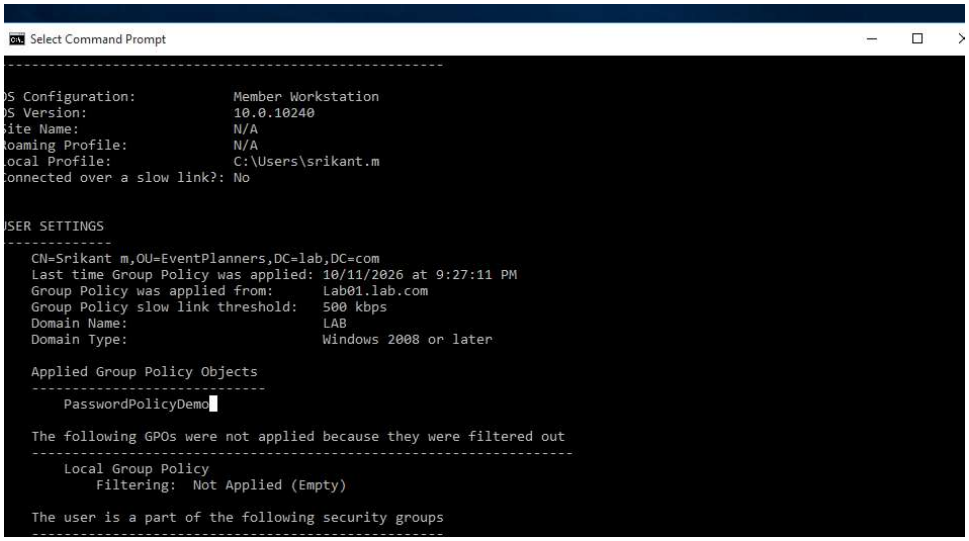


Figure 4: gpresult /r on the client: PasswordPolicyDemo under Applied Group Policy Objects

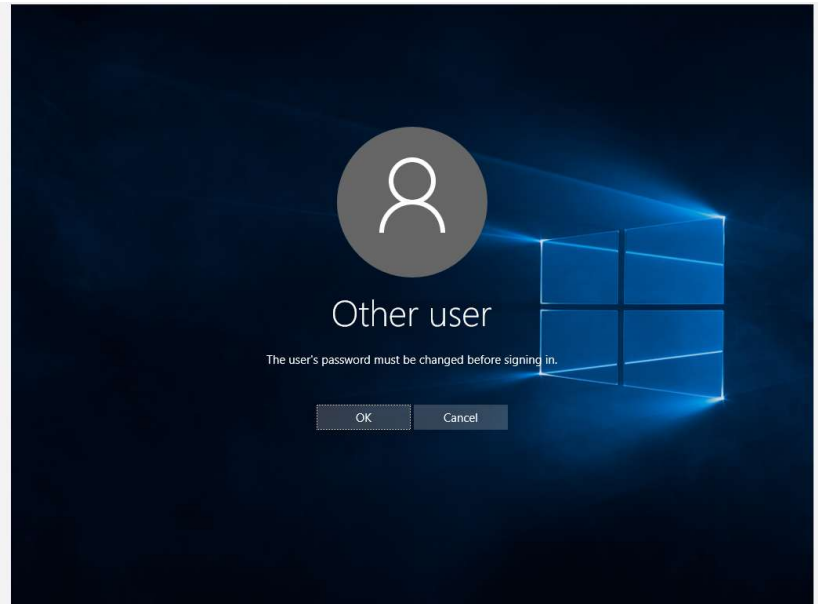


Figure 5: Sign-in message: the user's password must be changed before signing in

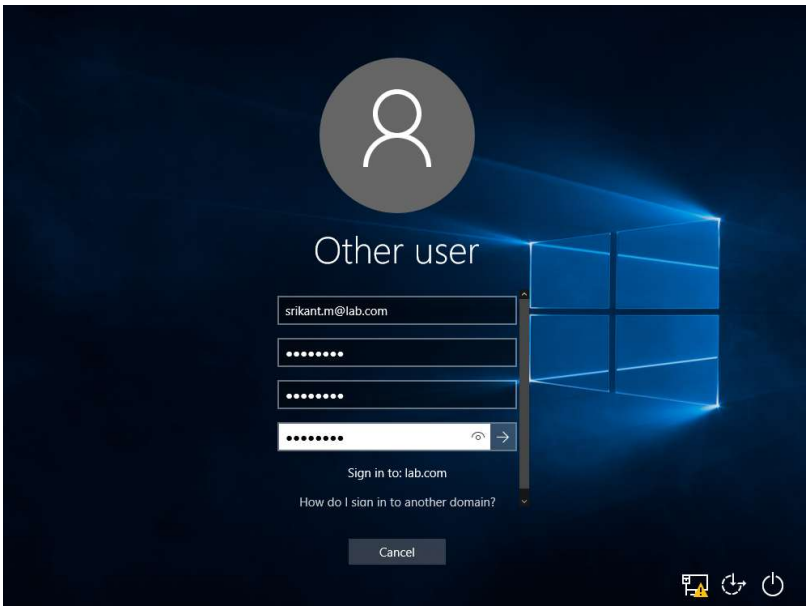


Figure 6: Password-change fields on the sign-in screen for srikant.m@lab.com

Task 2: Configure a GPO to restrict access to the Control Panel for all users in a specific OU, similar to how public computers in a cybercafe or college lab prevent users from changing system settings. Hint: In the GPO, navigate to User Configuration > Administrative Templates > Control Panel and set 'Prohibit access to Control Panel and PC settings' to Enabled.

Steps performed

1. In GPMC, created a GPO named Event Planners and linked it to the EventPlanners OU (Figure 7). The link is enabled, not enforced, and the GPO status is Enabled.
2. Edited the GPO and went to User Configuration > Administrative Templates > Control Panel.
3. Opened Prohibit access to Control Panel and PC settings and set it to Enabled, then clicked Apply and OK (Figure 8).
4. Signed in to the client PC as srikant.m (a user in the OU), pressed Windows + R, typed control and pressed Enter (Figure 9).

Result and verification

Instead of opening, Control Panel was blocked with a "Restrictions" message saying the operation was cancelled due to restrictions on the computer and to contact the system administrator (Figure 10). The policy works for users in the EventPlanners OU.

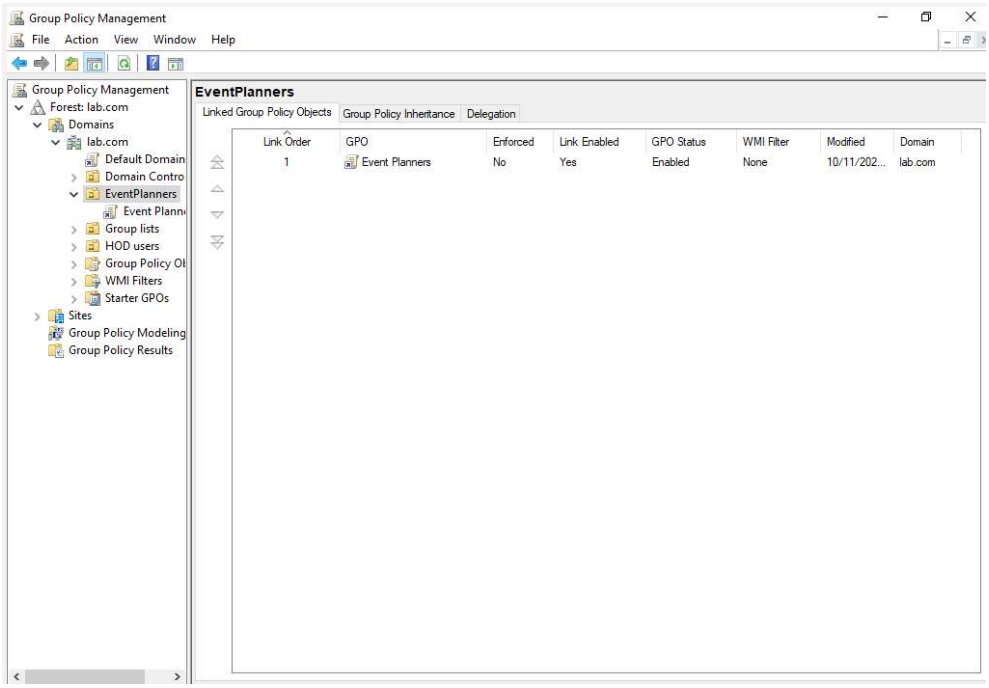


Figure 7: GPMC: the Event Planners GPO linked to the EventPlanners OU (link enabled, GPO enabled)

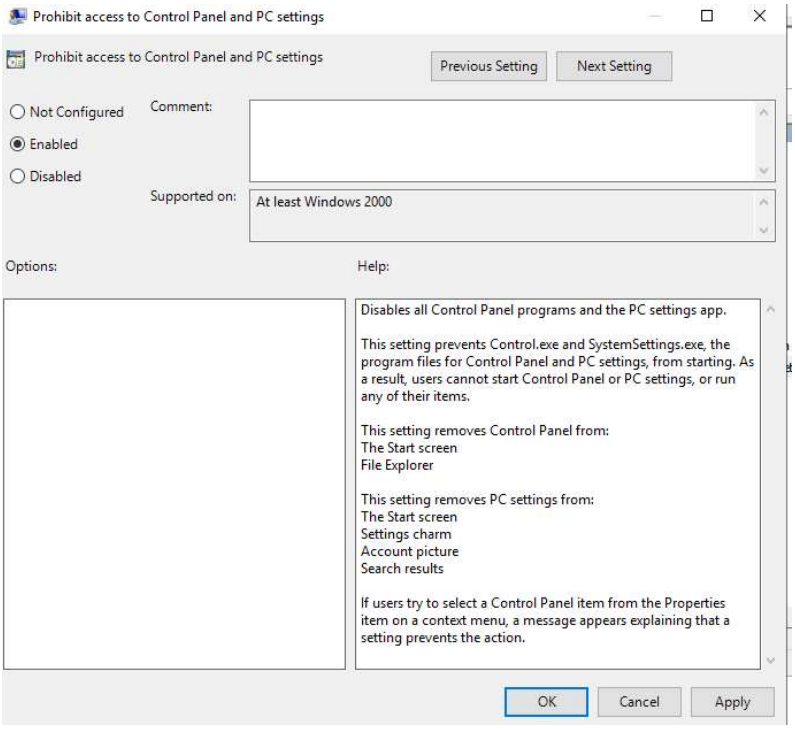


Figure 8: Prohibit access to Control Panel and PC settings set to Enabled

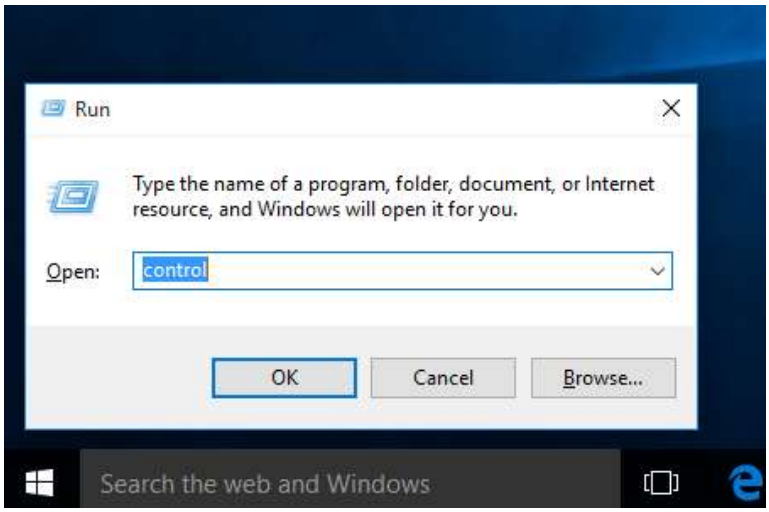


Figure 9: Run dialog on the client PC with the command control

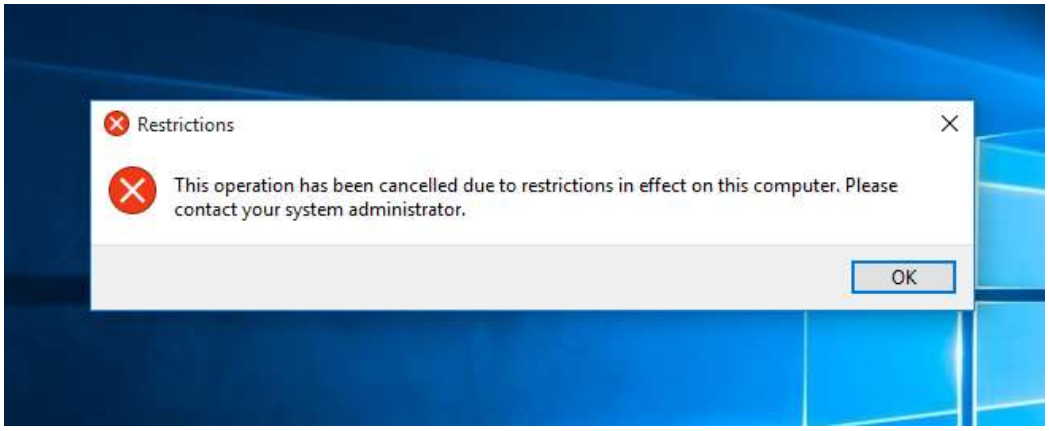


Figure 10: Result: "This operation has been cancelled due to restrictions in effect on this computer"

Task 3: Edit your 'PasswordPolicyDemo' GPO to enable account lockout after 3 invalid login attempts and set the lockout duration to 10 minutes. Run gpupdate /force on a client PC and test the lockout by entering the wrong password.

Steps performed

1. Edited PasswordPolicyDemo and went to Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Account Lockout Policy.
2. Set Account lockout threshold to 3 invalid logon attempts (Figure 11).
3. Set Account lockout duration to 10 minutes (Figure 12). Windows also set Reset account lockout counter after to 10 minutes and left Allow Administrator account lockout Enabled (Figure 13).
4. On the client PC ran gpupdate /force to apply the policy.
5. At the sign-in screen, entered a wrong password for srikant.m three times.

```
gpupdate /force
```

Verification

After the third wrong password, Windows displayed "The referenced account is currently locked out and may not be logged on to" for LAB\srikant.m (Figure 15). This confirms the lockout threshold works. The lockout test was run with the duration and reset counter set to 2 minutes (Figure 14) so the wait before the account unlocked was short; the required production values are 10 minutes (Figure 13).

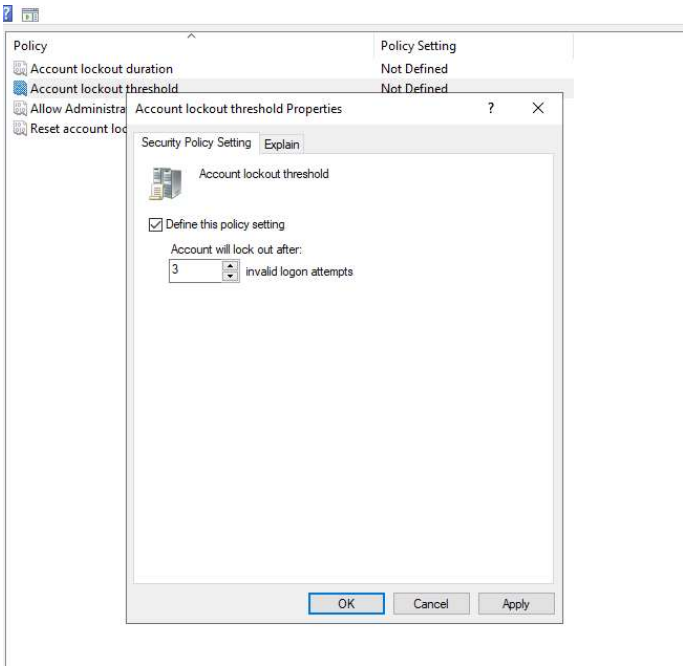


Figure 11: Account lockout threshold set to 3 invalid logon attempts

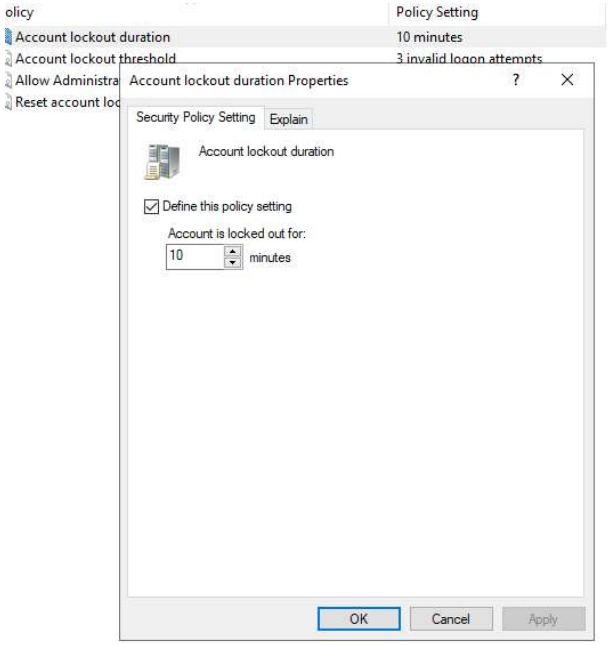


Figure 12: Account lockout duration set to 10 minutes

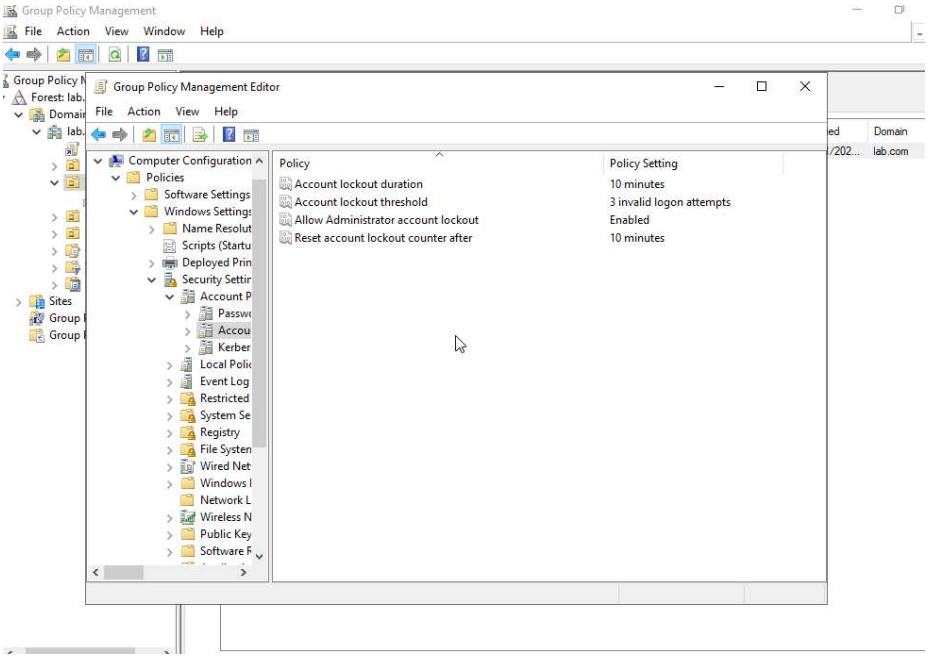


Figure 13: Account Lockout Policy in the GPO editor: 10 minutes, 3 attempts, 10 minutes reset counter

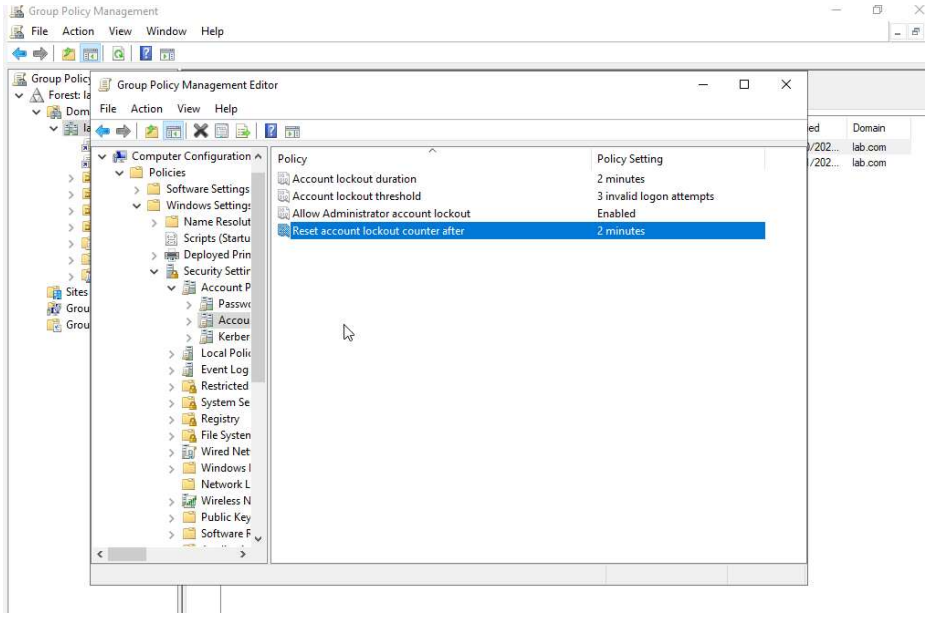


Figure 14: Lockout duration and reset counter shortened to 2 minutes for the lockout test

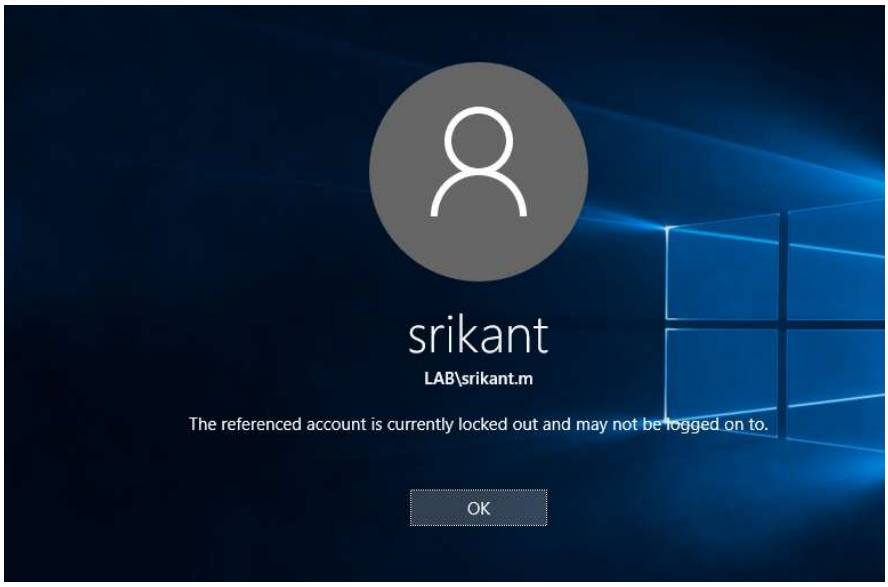


Figure 15: Account locked out after 3 wrong passwords

Task 4: Document all GPO settings you applied in this session (password policy, desktop restriction, lockout policy) in a table format, and for each setting, describe its expected impact on users in 1-2 sentences.

GPO	Setting and path	Value	Expected impact on users
PasswordPolicy Demo	Minimum password length. Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy	8 characters	Users cannot choose a password shorter than 8 characters. Any new or changed password that is too short is rejected, which makes passwords harder to guess.
Event Planners (linked to EventPlanners OU)	Prohibit access to Control Panel and PC settings. User Configuration > Administrative Templates > Control Panel	Enabled	Users in the OU cannot open Control Panel or PC Settings. Trying to start them shows a "Restrictions" message, so users cannot change system settings.
PasswordPolicy Demo	Account lockout threshold. Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Account Lockout Policy	3 invalid logon attempts	After 3 wrong passwords the account is locked. This blocks password-guessing attacks, but a user who mistypes repeatedly will be locked out too.
PasswordPolicy Demo	Account lockout duration (Account Lockout Policy)	10 minutes	A locked account stays locked for 10 minutes and then unlocks by itself, so a locked user can wait or ask an administrator to unlock it.
PasswordPolicy Demo	Reset account lockout counter after (Account Lockout Policy)	10 minutes	The count of failed attempts resets after 10 minutes, so occasional typos spread over time do not lead to a lockout.